MILITARY SPECIFICATION

LEAD, EXPLOSIVE

MARK 24 MOD 0

MARK 25 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of two types of explosive leads, referred to herein as the lead.

1.2 Classification. Leads covered by this specification shall be of the following types:

Type I. Lead, Explosive, MARK 24 MOD 0, Drawing 1137AS290.

Type II. Lead, Explosive, MARK 25 MOD 0, Drawing 1137AS292.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-21723 Composition CH-6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-C-45662	Calibration System Requirements.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL 1137AS290	Lead, Explosive, MARK 24 MOD 0.
DL 1137AS292	Lead, Explosive, MARK 25 MOD 0.
LD 533567	Detonator, Electric, MARK 87 MOD 0.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

3.1 Item definition. The Type I lead, when detonated by a MARK 87 MOD 0 Detonator, performs a transfer function to detonate a Type II lead in a safety-arming (S-A) device.

3.1.1 Government-furnished property. The Government will furnish the contractor with MARK 87 MOD 0 Detonators (LD 533567) in a quantity sufficient to perform the preproduction and production tests specified herein (see 6.2.1).

3.2 Characteristics.

3.2.1 Explosive charge. The explosive charge shall be Composition CH-6 in accordance with MIL-C-21723.

3.2.1.1 Moisture content. The moisture content of the explosive charge shall not exceed 0.20 percent at the time of loading.

3.2.2 Performance.

3.2.2.1 Function. The lead shall function when detonated by a MARK 87 MOD 0 Detonator.

3.2.2.2 Output. The output of the lead shall be evaluated by means of a steel dent test. The dent block shall be in accordance with Figure 1. The lead holder shall be in accordance with Figure 2. The leads and detonators, when confined in the appropriate lead holder and standing on the steel plate, shall produce, on explosion, a minimum average depth of dents as follows:

(a) Type I lead: Not less than 0.035 inch.

(b) Type II lead: Not less than 0.087 inch.

In addition, the depth of any single dent shall not be less than 0.030 inch from a Type I lead, and not less than 0.066 inch from a Type II lead.

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3.2.3 Radiographic. The lead shall be free from defects such as cracks, voids, fissures, or loose powder.

3.2.4 Environmental. The lead shall not be damaged, nor shall safety or subsequent performance be degraded, by exposure to any natural combination of service-use environments. These shall include, but not be limited to, the following environments.

3.2.4.1 Aircraft vibration (nondestructive). The lead shall meet the requirements of MIL-STD-810, Method 514.2, Category d.1, Procedure IIA, and shall function with the proper output after subjection to the environment of 4.5.1 (see 3.2.2).

3.2.4.2 Jolt (destructive). The lead shall meet the requirements of MIL-STD-331, Test 101.2, Procedure II, and shall be safe to handle and dispose of after subjection to the environment of 4.5.2.

3.2.4.3 Forty-foot drop (destructive). The lead shall meet the requirements of MIL-STD-331, Test 103, Procedures 1 and 2, and shall be safe to handle and dispose of after subjection to the environment of 4.5.3.

3.2.4.4 Temperature and humidity (nondestructive). The lead shall meet the requirements of MIL-STD-331, Test 105.1, Single Chamber-Cycling Method, and shall function with the proper after subjection to the environment of 4.5.4 (see 3.2.2).

3.2.4.5 Transportation vibration-temperature (nondestructive). The lead shall meet the requirements of MIL-STD-331, Test 119, Procedure I, and shall function with the proper output after subjection to the environment of 4.5.5 (see 3.2.2).

3.3 Design and construction.

3.3.1 Production drawings. The lead shall be manufactured and assembled in accordance with the drawings and documents listed on DL 1137AS290 (Type I) and DL 1137AS292 (Type II) and as specified herein.

3.3.2 Test equipment calibration. When specified in the contract or purchase order, test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. The calibration plan shall be submitted to the Government for approval prior to use (see 6.2.2).

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3.3.3 Test plan and test procedures. When specified in the contract or purchase order and using MIL-STD-831 as a guide, the contractor shall prepare and submit to the Government for approval, a test plan and test procedures prior to testing (see 6.2.2).

3.4 First article.

3.4.1 Preproduction sample. Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample of 50 leads to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

3.5 Workmanship. The lead shall be constructed and finished in such a manner as to ensure compliance with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspections (see 4.1.2.1).

(b) Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspections. The preproduction sample of 50 leads shall be subdivided and subjected to the tests and inspections of Figure 3 (see 6.2.1). The rejection of any sample lead shall be cause for rejection of the preproduction sample.

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4.2 Quality conformance inspections. A quality conformance sample of 50 leads, randomly selected from the production lot in accordance with MIL-STD-105, Normal Sampling Plan, Inspection Level I, AQL of 0.65, shall be subjected to the tests and inspections of Figure 4.

4.2.1 Inspection lot. The term inspection lot, as applied to Government inspection of leads, shall mean a collection of leads submitted by the contractor for Government inspection.

4.2.2 Production lot. Unless otherwise specified in the contract or purchase order, a production lot shall consist of 1,200 to 3,200 leads including the leads required for test purposes (see 6.2.1).

4.2.3 Production inspections. The mandatory inspection requirements for the lead are specified by the classification of characteristics on the referenced drawing and also, if required, by supplemental quality assurance provisions in the contract or purchase order (see 6.2.1).

4.3 Test conditions and provisions.

4.3.1 Test conditions. Unless otherwise specified herein, the test methods of 4.4 shall be performed under the following conditions:

(a)	Temperature:	Room ambient of 65 ±5 degrees Fahrenheit (°F) to 95 ±5°F.
(b)	Altitude:	Normal ground elevation.
(c)	Humidity:	Room ambient to 70 percent relative humidity maximum.

4.3.2 Test and inspection equipment and facilities. The contractor shall furnish and maintain all necessary test equipment, facilities, and personnel necessary to perform all acceptance tests. The test equipment shall be of sufficient accuracy and quality to permit satisfactory performance of the required tests and examinations.

4.3.3 Test equipment calibration. Test equipment shall be calibrated and maintained as specified in 3.3.2.

4.3.4 Test plan and test procedures. The contractor shall prepare a test plan and test procedures as specified in 3.3.3.

4.3.5 Gages. The contractor shall provide whatever gages are necessary and adequate to assure that the material to which this specification applies shall meet the dimensional requirements shown on the applicable drawings. When specified in the contract or purchase order, the Government will furnish drawings of pertinent Navy final inspection gages (see 6.2.1).

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4.4 Methods of examination and test.

4.4.1 Explosive charge moisture content. The moisture content of the explosive charge shall be measured as specified herein on each day of production or on each lot, whichever requires the more frequent testing, plus such additional tests as the cognizant inspector considers warranted by variations in humidity, manufacturing process, or materials. An approximate 2.0-gram sample of the material under investigation shall be collected in a tared weighing bottle, having a cap or stopper ground to fit, at the site of and during the production of the lot. The stoppered bottle and contents shall be accurately weighed. After weighing, the stopper shall be removed, and the open bottle and contents shall be dried at 212.0 ±2.7°F for a period of 40 to 50 minutes, cooled in a desiccator, stoppered, and reweighed on the same balance that was used for the first measurement. The loss in weight of the stoppered bottle and contents shall be calculated as the percentage of moisture in the sample. To be acceptable, the charge shall meet the requirements of 3.2.1.1. Nonconforming compositions shall be rejected and discarded.

CAUTION

The function and output tests involve hazardous operations. Proper safety precautions shall be taken (see 6.3).

4.4.2 Function. The lead shall be detonated in an output test assembly as shown on Figure 5 (Type I) and Figure 6 (Type II). To be acceptable, the lead shall meet the requirements of 3.2.2.1.

4.4.3 Output. The lead shall be detonated in an output test assembly as shown on Figure 5 (Type I) and Figure 6 (Type II). Measurements of the dent block shall be in accordance with Figure 7. To be acceptable, the lead shall meet the requirements of 3.2.2.2.

4.4.4 Radiographic. The lead shall be radiographically inspected in accordance with MIL-STD-453 for any evidence of cracks, voids, fissures, or loose powder. To be acceptable, the lead shall meet the requirements of 3.2.3.

4.5 Environmental.

4.5.1 Aircraft vibration (nondestructive). The lead shall be tested in accordance with MIL-STD-810, Method 514.2, Category d.1, Procedure IIA. To be acceptable, the lead shall meet the requirements of 3.2.4.1.

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4.5.2 Jolt (destructive). The lead, which has been subjected to the test of 4.4.4, shall be tested in accordance with MIL-STD-331, Test 101.2, Procedure II. To be acceptable, the lead shall meet the requirements of 3.2.4.2.

4.5.3 Forty-foot drop (destructive). The lead, which has been subjected to the test of 4.4.4, and is installed in an S-A device or a simulated configuration, shall be tested in accordance with MIL-STD-331, Test 103, Procedures 1 and 2. To be acceptable, the lead shall meet the requirements of 3.2.4.3.

4.5.4 Temperature and humidity (nondestructive). The lead shall be tested in accordance with MIL-STD-331, Test 105.1, Single Chamber-Cycling Method, except the test shall be conducted for one 14-day cycle. To be acceptable, the lead shall meet the requirements of 3.2.4.4.

4.5.5 Transportation vibration-temperature (nondestructive). The lead shall be tested in accordance with MIL-STD-331, Test 119, Procedure I. To be acceptable, the lead shall meet the requirements of 3.2.4.5.

4.6 Packaging, packing, and marking inspection. Prior to shipment, the packaging, packing, and marking of the lead shall be inspected to ensure conformance to the requirements of Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102.

5.1.1 Level C. Leads shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102.

5.2.1 Level C. Leads, packaged as described in 5.1.1, shall be packed in a manner which will provide protection from damage during direct shipment from the supply source to the first receiving activity. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2.1), all marking shall be in accordance with MIL-STD-129, and Code of Federal Regulations 49 CFR 171-178.

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6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The lead is intended for use in an S-A device for guided missiles.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Number, title, and date of this specification.

(b) Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

(c) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 50 leads (either Type I or Type II) as specified in 3.4.1 and 4.1.2.1. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

(d) Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

(e) Production lot, if other than as specified in 4.2.2.

(f) Supplemental quality assurance provisions (see 4.2.3).

(g) Government final inspection gage drawings (see 4.3.5).

(h) Special marking (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423

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incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.3.2	Calibration Plan	UDI-T-23742A
3.3.3	Test Plan and Test Procedures	DI-T-5204

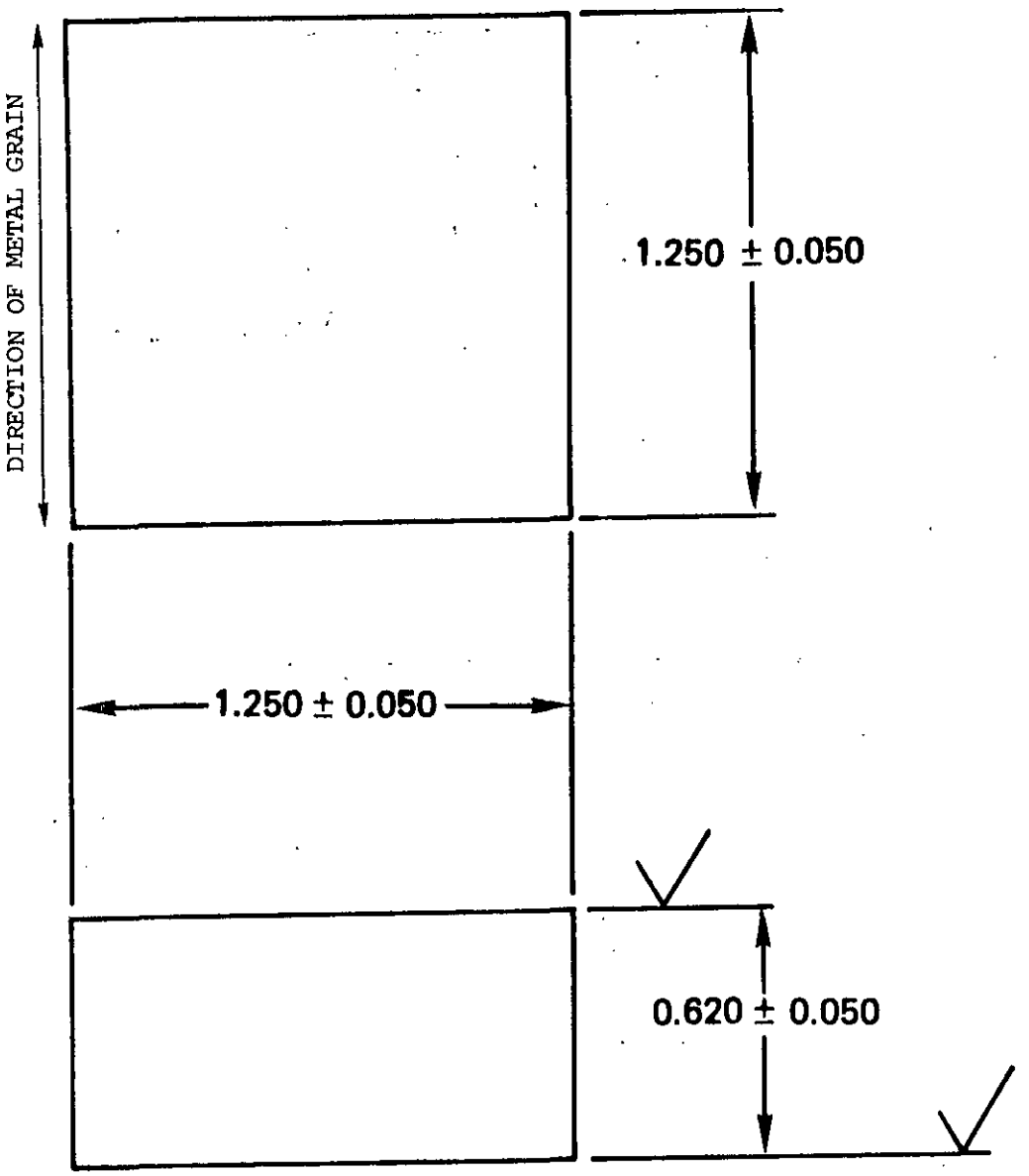
(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DoD Instruction 4145.26M, DoD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

Preparing activity:
Navy-AS

(Project Number 1336-N300)

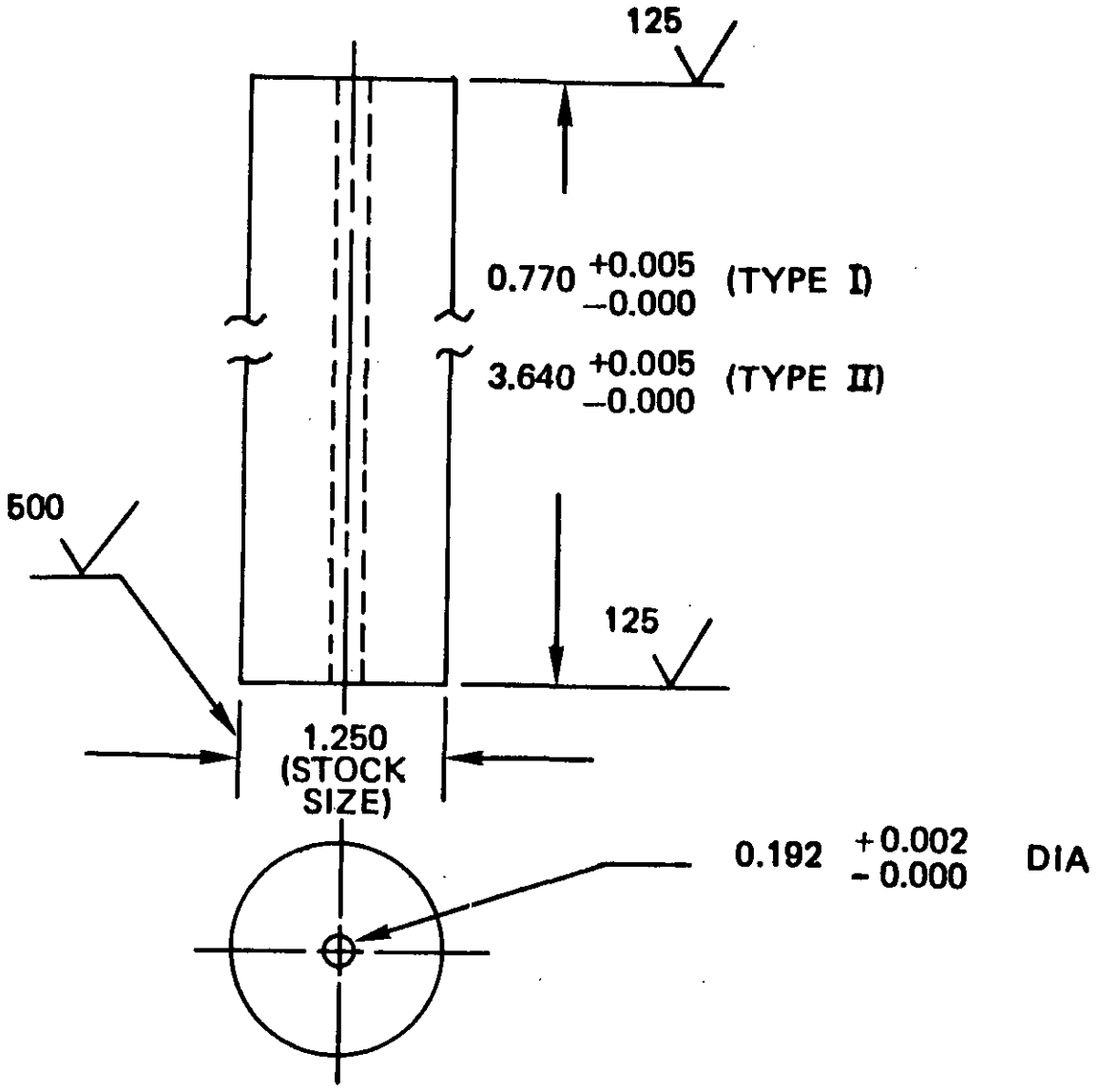
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NOTE:
Material: Steel, SAE 1018 or 1020, cold-rolled, Rockwell hardness B70 to B85. Finish (√) to be as rolled.

FIGURE 1. Dent block.

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NOTES:
1. All dimensions ±0.005, unless otherwise specified.
2. Material: Steel, SAE 1018 through 1035, cold- or hot-rolled.

FIGURE 2. Detonator and lead holder (Type I and Type II).

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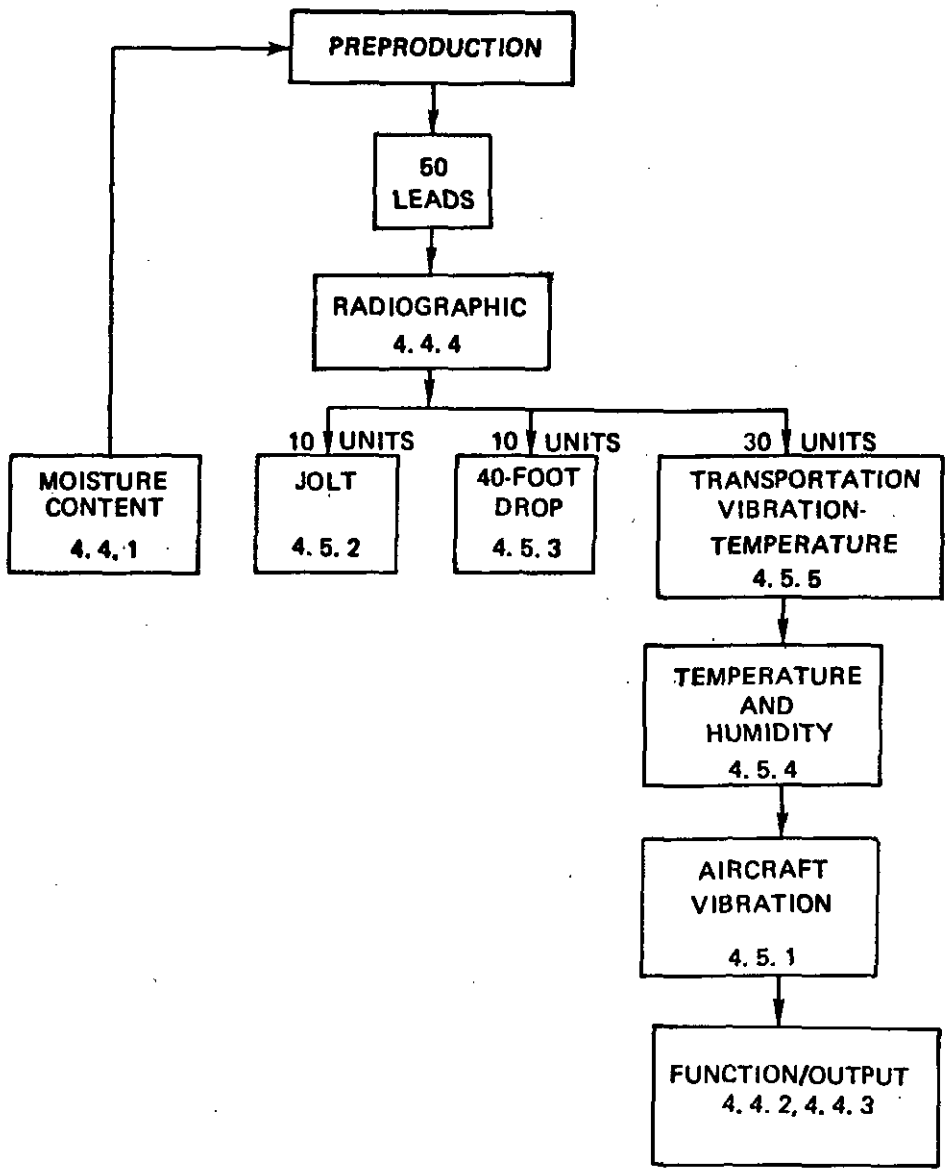


FIGURE 3. Preproduction sample test sequence.

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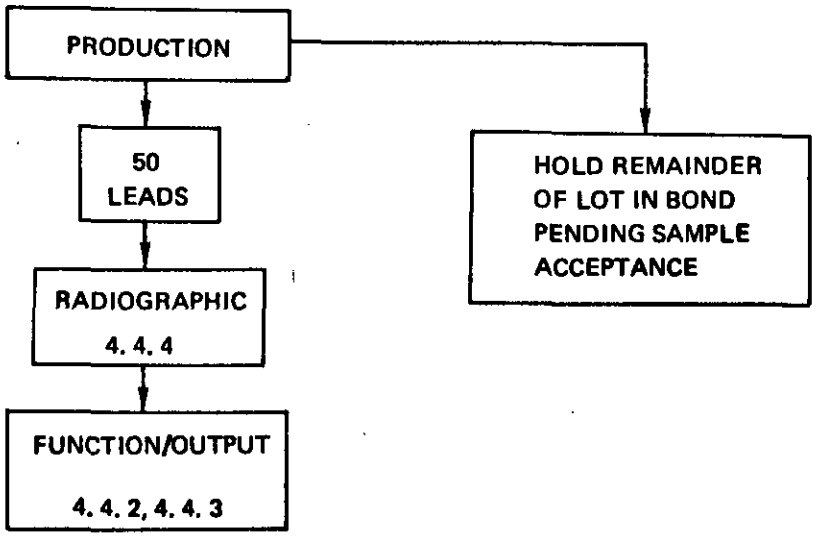


FIGURE 4. Production sample test sequence.

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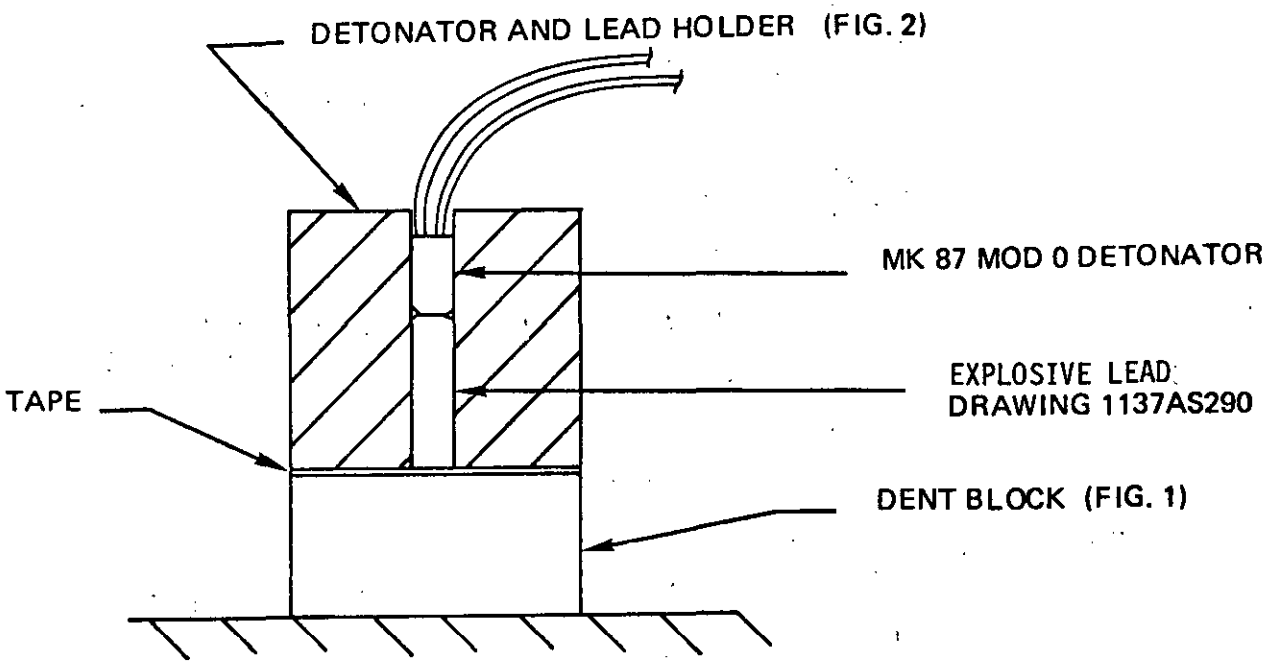


FIGURE 5. Output test assembly (Type I).

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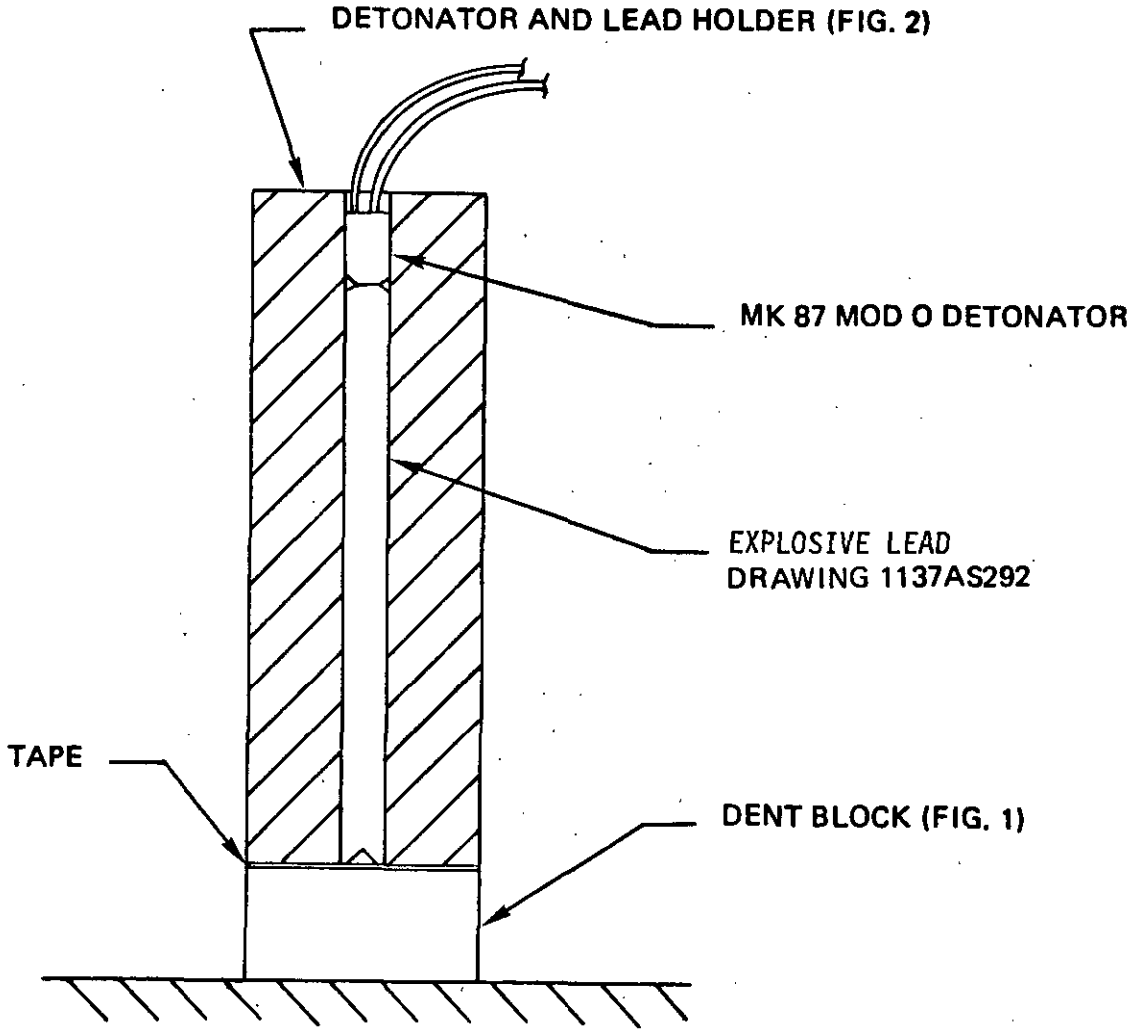


FIGURE 6. Output test assembly (Type II).

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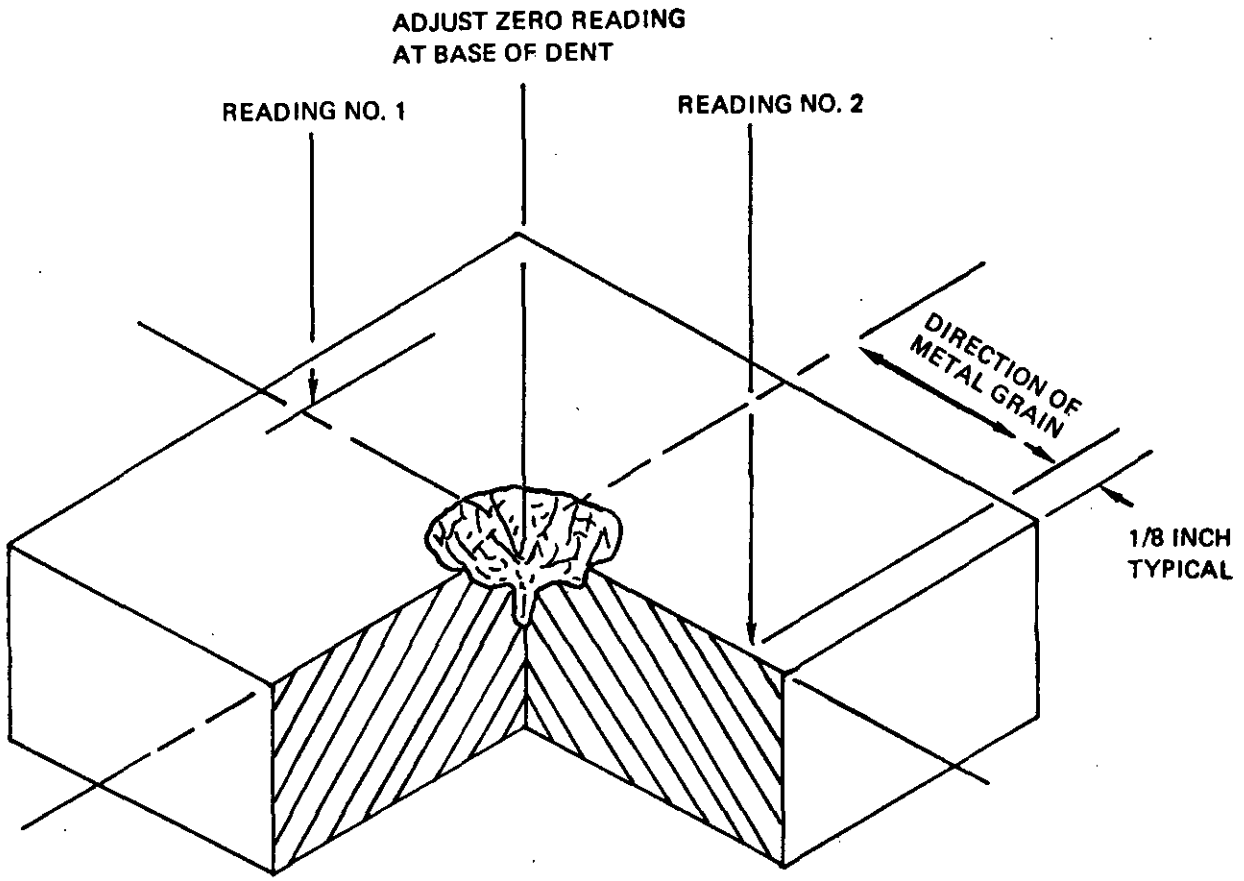


FIGURE 7. Locations for measuring depth of dent in steel block (Type I and Type II).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-L-85297(AS), Lead, Explosive, MARK 24 MOD 0, MARK 25 MOD 0

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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C. REASON FOR RECOMMENDED CHANGE(S)

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SUBMITTED BY *(Printed or typed name and address — Optional)*

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DATE

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